

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Mathematics — SSC-I (9th Grade)

Syllabus: Chapter 9 — Geometry and Polygons | ANSWER KEY (Version 1)

Total Marks: 65

Time Allowed: 2h 30m

Version: 1

SECTION A — MCQ Answer Key (12 × 1 = 12 Marks)

Q1. C — Axiom

An **axiom** is assumed true without proof (e.g., "a number equals itself"). A theorem requires proof; a conjecture needs verification.

Q2. B — Conjecture

A **conjecture** is believed true but unproven. Axioms are accepted without proof; theorems are already proved.

Q3. B — ~

The **similarity symbol** is $\sim$. The congruence symbol $\cong$ means same shape and size. Similarity means same shape only.

Q4. B — k^2

Ratio of **areas** = (ratio of corresponding lengths) 2 = k^2 . Common error: choosing k (linear ratio) or k^3 (volume ratio).

Q5. C — $(n-2) \times 180^\circ$

Formula comes from splitting an n -gon into $(n-2)$ triangles. For $n=4$: $(4-2) \times 180 = 360^\circ$ (correct for quadrilateral).

Q6. C — 360°

Sum of **exterior angles** of any convex polygon is always 360° , regardless of the number of sides.

Q7. B — 120°

Regular hexagon: $n=6$. Interior angle = $(6-2) \times 180 / 6 = 720 / 6 = 120^\circ$. Option A (108°) belongs to a regular pentagon.

Q8. B — $\frac{1}{2} P \times a$

Area of regular polygon = $\frac{1}{2} \times \text{Perimeter} \times \text{apothem}$. Apothem is the perpendicular from centre to any side.

Q9. A — 3 : 2

$V_1/V_2 = k^3$, so $k^3 = 27/8$, thus $k = 3/2$. Ratio of **lengths** = 3 : 2. Option B (9:4) is the area ratio, not length.

Q10. C — Perpendicular bisector of AB

The **perpendicular bisector** of segment AB is the locus of all points equidistant from A and B. Every point on it satisfies $PA = PB$.

Q11. B — 8

$n = 360^\circ / \text{exterior angle} = 360/45 = 8$ sides (regular octagon). Common error: dividing 180 by 45 giving 4.

Q12. B — $n(n-3)/2$

Each vertex connects to $(n-3)$ non-adjacent vertices. Total unique diagonals = $n(n-3)/2$. Option A counts all vertex pairs including sides.

SECTION B — Short Questions (11 × 3 = 33 Marks)

2(i) — Axiom vs Conjecture 3 marks**Axiom:** A mathematical statement assumed to be true without any proof. 1 markExample: "If $a = b$ and $b = c$, then $a = c$ " (Transitive property). Axioms are also called postulates when related to geometry.**Conjecture:** A statement believed to be true but whose truth has not yet been proved. 1 mark

Example: From the pattern 4, 8, 12, 16, ... one conjectures the next number is 20 (adding 4). This is plausible but not yet formally proved.

Key difference: An axiom is accepted without proof; a conjecture still requires verification or proof.

1 mark

OR**Theorem and Its Six Steps** 3 marks**Theorem:** A mathematical statement that can be proved or supposed provable through logical reasoning.

1 mark

Six steps of geometrical proof 2 marks (any 6 correctly named):

1. **Statement** — describe the theorem in words.
2. **Figure** — draw a neat labelled diagram.
3. **Given** — write given information symbolically.
4. **Required to Prove** — state the result to be proved symbolically.
5. **Construction** — add any extra lines/points needed.
6. **Proof** — write statements with reasons step by step.

2(ii) — Deductive Proof and Inductive Verification 3 marks**Deductive proof:** 2 marksLHS = $3x + 9 = 3(x + 3)$ [factoring out 3] = RHS. Hence proved for all real x .**Inductive verification for $x = 2$:** 1 markLHS = $3(2) + 9 = 6 + 9 = 15$. RHS = $3(2 + 3) = 3(5) = 15$. LHS = RHS. ✓**OR****Computing $(103)^2$ by Deductive Approach** 3 marks**General identity (deductive):** $(a + b)^2 = a^2 + 2ab + b^2$ 1 mark**Particular case:** Write $103 = 100 + 3$, so $a = 100$, $b = 3$. 1 mark $(103)^2 = (100)^2 + 2(100)(3) + (3)^2 = 10000 + 600 + 9 = 10609$. 1 mark**2(iii) — Proving $\triangle ABC \sim \triangle DEF$** 3 marksGiven: $\angle A = \angle D = 40^\circ$, $AC/DF = AB/DE = 2/3$. 1 markSince two pairs of corresponding sides are proportional ($AC/DF = AB/DE$) and the included angle between those sides is equal ($\angle A = \angle D$), by the **SAS similarity condition**, $\triangle ABC \sim \triangle DEF$. 2 marks**OR****Finding Sides of $\triangle XYZ$** 3 marksGiven: $\triangle PQR \sim \triangle XYZ$ with ratio $PQ/XY = QR/YZ = PR/XZ = 2$. 1 mark $XY = PQ/2 = 6/2 = 3 \text{ cm}$. $YZ = QR/2 = 8/2 = 4 \text{ cm}$. $XZ = PR/2 = 4/2 = 2 \text{ cm}$. 2 marks

2(iv) — Finding x and y in Similar Triangles 3 marks

Since $\triangle LMN \sim \triangle XYZ$: $LM/XY = LN/XZ = MN/YZ$. 1 mark

$MN/YZ = 4/2 = 2$ (scale factor). 1 mark

$x/3 = 2 \Rightarrow x = 6 \text{ cm}$. $6/y = 2 \Rightarrow y = 3 \text{ cm}$. 1 mark

OR**Height of Wall using Similar Triangles** 3 marks

Pole height = 7.5 m, pole shadow = 15 m; wall shadow = 45 m. 1 mark

By similar triangles: pole height / pole shadow = wall height / wall shadow. 1 mark

$7.5/15 = h/45 \Rightarrow h = (7.5 \times 45)/15 = 22.5 \text{ m}$. 1 mark

2(v) — Area Ratio of Similar Rectangles 3 marks

General rule: $A_1/A_2 = (l_1/l_2)^2 = k^2$. 1 mark

Here $k = 1/2$, so $A_1/A_2 = (1/2)^2 = 1/4$. 1 mark

$A_2 = 4 \times A_1 = 4 \times 6 = 24 \text{ cm}^2$. 1 mark

OR**Finding Corresponding Side from Area Ratio** 3 marks

$A_1/A_2 = (l_1/l_2)^2 \Rightarrow 25/108 = (2/l_2)^2$. 1 mark

$(l_2/2)^2 = 108/25 \Rightarrow l_2/2 = \sqrt{(108/25)} = \sqrt{(4.32)}$. 1 mark

$l_2 = 2 \times \sqrt{(4.32)} \approx 2 \times 2.078 = 6 \text{ cm}$ (approx). 1 mark

(Exact: $A_1/A_2 = (l_1/l_2)^2 \Rightarrow l_2 = l_1 \times \sqrt{(A_2/A_1)} = 2\sqrt{(108/25)} = 2 \times 6\sqrt{3}/5 = 12\sqrt{3}/5 \text{ cm}$.)

2(vi) — Ratios for Similar Cylinders (r = 3 cm, R = 5 cm) 3 marks

Scale factor $k = r/R = 3/5$. 1 mark

(a) Ratio of curved surface areas = $k^2 = (3/5)^2 = 9 : 25$. 1 mark

(b) Ratio of volumes = $k^3 = (3/5)^3 = 27 : 125$. 1 mark

OR**Volume of Actual Bus (Scale 1 : 10)** 3 marks

Scale factor $k = 1/10$ (model to actual). 1 mark

$V_{\text{model}}/V_{\text{actual}} = k^3 = (1/10)^3 = 1/1000$. 1 mark

$V_{\text{actual}} = 30 \times 1000 = 30\,000 \text{ m}^3$. 1 mark

2(vii) — Regular and Irregular Polygons 3 marks

Regular polygon: A polygon in which all sides and all interior angles are equal. Also called equilateral and equiangular. Examples: equilateral triangle, square. 1 mark

Irregular polygon: A polygon in which at least one side or one angle is not equal to the others. Examples: scalene triangle, rectangle (equal angles but unequal sides). 1 mark

Each interior angle of a regular n-gon = $(n - 2) \times 180^\circ / n$. 1 mark

OR**Polygon from Exterior Angle 120°** 3 marks

Sum of exterior angles = 360° . Number of sides = $360/120 = 3 \Rightarrow$ **Equilateral triangle**. 2 marks

Interior angle = $180^\circ - 120^\circ = 60^\circ$. 1 mark

2(viii) — Interior Angles of 9-Sided Polygon 3 marks

$$\text{Sum} = (n - 2) \times 180^\circ = (9 - 2) \times 180^\circ = 7 \times 180^\circ = \mathbf{1260^\circ}.$$
 2 marks

$$\text{Each interior angle (regular nonagon)} = 1260/9 = \mathbf{140^\circ}.$$
 1 mark

OR**Finding n when Sum = 2880°** 3 marks

$$(n - 2) \times 180 = 2880$$
 1 mark

$$n - 2 = 2880/180 = 16 \Rightarrow n = \mathbf{18}.$$
 1 mark

An 18-sided polygon is called an **octadecagon**. 1 mark**2(ix) — Exterior Angles of Pentagon** 3 marks

$$\text{Sum of exterior angles} = 360^\circ.$$
 1 mark

$$(y+5) + (2y+3) + (3y+2) + (4y+1) + (5y+4) = 360$$

$$15y + 15 = 360 \Rightarrow 15y = 345 \Rightarrow y = \mathbf{23}.$$
 1 mark

$$\text{Angles: } 28^\circ, 49^\circ, 71^\circ, 93^\circ, 119^\circ \text{ (check: } 28+49+71+93+119 = 360^\circ \checkmark).$$
 1 mark

OR**Interior = 5 × Exterior: Finding n** 3 marks

$$\text{Let exterior angle} = e, \text{ interior angle} = 5e. \text{ Interior} + \text{exterior} = 180^\circ \Rightarrow 6e = 180^\circ \Rightarrow e = 30^\circ.$$
 2 marks

$$n = 360/30 = \mathbf{12} \text{ sides (regular dodecagon).}$$
 1 mark

2(x) — Regular Hexagon Plot (side 10 m, apothem 8.66 m) 3 marks

$$\text{(a) Perimeter} = n \times s = 6 \times 10 = \mathbf{60 \text{ m.}}$$
 1 mark

$$\text{(b) Area} = \frac{1}{2} \times P \times a = \frac{1}{2} \times 60 \times 8.66 = \mathbf{259.8 \text{ m}^2}.$$
 2 marks

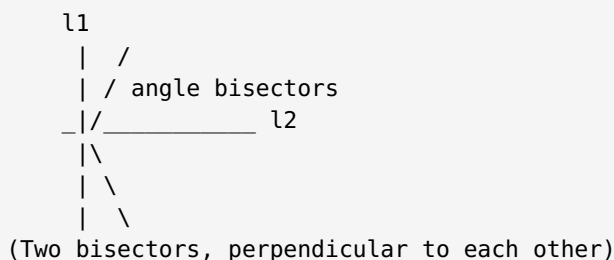
OR**Equilateral Triangle Lawn (side 5 m)** 3 marks

$$\text{Perimeter} = 3 \times 5 = \mathbf{15 \text{ m.}}$$
 1 mark

$$\text{Cost} = 15 \times 220 = \mathbf{\text{Rs. } 3,300}.$$
 2 marks

2(xi) — Locus Equidistant from Two Intersecting Lines 3 marks**Locus:** The set of all points that satisfy a given geometrical condition; a path traced by a moving point.

1 mark

Locus description: The locus of a point equidistant from two intersecting lines is a pair of straight lines (the **angle bisectors**) of the angles formed by the two lines. 1 mark

Sketch shows two lines intersecting at O. The locus consists of two lines through O bisecting the four angles at O. 1 mark

OR**Locus of P at Distance 4 cm from Line l** 3 marks**Description:** The locus is a pair of straight lines, one on each side of l, each parallel to l and at a perpendicular distance of 4 cm from l. 2 marks**Two loci exist** (one on each side of the line). The point Q moving at 8 cm from l is NOT on this locus.

1 mark

SECTION C — Long Questions (4 × 5 = 20 Marks)

3(i) — Reasoning Types and Algebraic Proof 5 marks

(a) Inductive Reasoning: Working from specific observations to a general result. It is not always logically valid. 1 mark

Example: Every day school starts at 8:00 am and you leave at 7:15 am and arrive on time. You conclude leaving at 7:15 always gets you there on time.

Deductive Reasoning: Working from a general rule to a specific conclusion. It is logically reliable.

1 mark

Example: All numbers equal themselves (axiom). Therefore $5 = 5$.

(b) Inductive verification for $(x+1)^2 + 5 = x^2 + 2x + 6$: 1 mark

$x = 2$: LHS = $(3)^2 + 5 = 14$; RHS = $4 + 4 + 6 = 14$ ✓

$x = 5$: LHS = $(6)^2 + 5 = 41$; RHS = $25 + 10 + 6 = 41$ ✓

$x = 10$: LHS = $(11)^2 + 5 = 126$; RHS = $100 + 20 + 6 = 126$ ✓

Deductive proof (general): 2 marks

LHS = $(x+1)^2 + 5 = x^2 + 2x + 1 + 5 = x^2 + 2x + 6 = \text{RHS}$. Hence proved for all real x .

OR

Definitions: Statement, Proposition, Corollary, Converse 5 marks

(a) Mathematical statement: A meaningful composition of words that can be either true or false; may contain words and symbols. 1 mark

Example: "Sum of 2 and 5 is 7" (true statement).

Proposition and its three parts 1 mark: (i) *Premise* (assumption), (ii) *Argument* (logical chain from premise to conclusion), (iii) *Conclusion* (result after applying the argument). The conclusion must be true when premises are true and argument is valid.

Corollary: A result that can be deduced directly from an already proved theorem, without much additional work. 1 mark

Converse of a theorem: Formed by interchanging the "given" and "to prove" parts. The converse is not necessarily a theorem. 1 mark

(b) Mathematical statements: 1 mark

(i) $34 + 16 \neq 50$: **True mathematical statement** (it is false as $34+16=50$, so the $\neq$ makes it false, but it is still a statement).

(ii) $(a+b)^2 = a^2 + 2ab + b^2$: **True mathematical statement.**

(iii) $xy + z = 12$: **Open sentence** (not a mathematical statement since values of x, y, z are unknown and truth value cannot be determined).

3(ii) — DE ∥ BC, Finding Ratios and Areas 5 marks

(a) In $\triangle ABC$, $DE \parallel BC$ with D on AB and E on AC. By the **Basic Proportionality Theorem** (Thales' theorem), a line parallel to one side of a triangle divides the other two sides proportionally. 2 marks
 Therefore: $AD/DB = AE/EC$. Here $AD = 4$ cm, $DB = 6$ cm, so $AD/DB = 4/6 = 2/3$. ✓

(b) $AE/EC = 2/3$. Since $AC = AE + EC = 15$ cm: 1 mark
 $AE/(15 - AE) = 2/3 \Rightarrow 3 AE = 30 - 2 AE \Rightarrow 5 AE = 30 \Rightarrow AE = 6$ cm.

(c) $AD/AB = 4/(4+6) = 4/10 = 2/5$. 2 marks
 Area of $\triangle ADE$ / Area of $\triangle ABC = (AD/AB)^2 = (2/5)^2 = 4/25$.
 (If Area of $\triangle ABC = 256$ cm², then Area $\triangle ADE = (4/25) \times 256 \approx 40.96$ cm².)

OR**Shadow and Pole Problem (Similar Triangles)** 5 marks

Let Wajid's height = h and his shadow = $(2/3)h$. Sun angle is the same for Wajid and the pole, so the two triangles are similar (**AA similarity**). 1 mark

(a) Wajid's shadow / Wajid's height = Pole's shadow / Pole's height 2 marks
 $(2/3) = 12 / \text{Pole height} \Rightarrow \text{Pole height} = 12 / (2/3) = 12 \times 3/2 = 18$ m.

(b) Wajid's height = 1.5 m. Shadow = $(2/3) \times 1.5 = 1$ m. 1 mark

(c) Yes, the triangles are similar by **AA similarity**: both share the same sun angle (angle of elevation) and both have a right angle at ground level. 1 mark

3(iii) — Similar Triangles ADE and ABC with Areas 5 marks

(a) In $\triangle ADE$ and $\triangle ABC$: $\angle A$ is common. Since $DE \parallel BC$, corresponding angles $\angle ADE = \angle ABC$ and $\angle AED = \angle ACB$. By **AA similarity**, $\triangle ADE \sim \triangle ABC$. 2 marks

(b) $DE/BC = AD/AB = 5/8$. 1 mark

(c)(i) Area $\triangle ADE$ / Area $\triangle ABC = (AD/AB)^2 = (5/8)^2 = 25/64$. 1 mark
 Area $\triangle ADE = (25/64) \times 256 = 100$ cm².

(c)(ii) Area of trapezium DBCE = Area $\triangle ABC$ - Area $\triangle ADE = 256 - 100 = 156$ cm². 1 mark

OR**Similar Cylinders P and Q (r = 3 cm, R = 5 cm)** 5 marks

Scale factor $k = r/R = 3/5$. 1 mark

(a) $V_P/V_Q = k^3 = (3/5)^3 = 27/125$ (ratio 27 : 125). 1 mark

(b) $162/V_Q = 27/125 \Rightarrow V_Q = (162 \times 125)/27 = 750$ cm³. 2 marks

(c) $A_P/A_Q = k^2 = (3/5)^2 = 9/25$. $A_P/250 = 9/25 \Rightarrow A_P = (9/25) \times 250 = 90$ cm². 1 mark

3(iv) — Interior Angles, Unknown Angle, and Exterior Angle 5 marks**(a)** $n = 13$: $\text{Sum} = (13 - 2) \times 180 = 11 \times 180 = \mathbf{1980^\circ}$. 1 mark

"9 right angles" $= 9 \times 90 = 810^\circ$. Check: $(n-2) \times 180 = 810 \Rightarrow n-2 = 4.5 \Rightarrow n = 6.5$. Since n must be a whole number, **it is not possible**. 1 mark

(b) Hexagon: $\text{sum} = (6-2) \times 180 = 720^\circ$. 1 mark

$$90 + 155 + 112 + x + 140 + 98 = 720$$

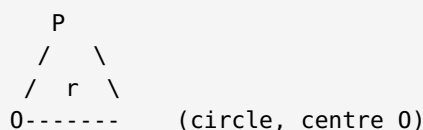
$$595 + x = 720 \Rightarrow x = \mathbf{125^\circ}$$

(c) Each exterior angle $= 12^\circ$. $n = 360/12 = \mathbf{30 \text{ sides}}$. 1 mark

$$\text{Sum of interior angles} = (30-2) \times 180 = 28 \times 180 = \mathbf{5040^\circ}$$
. 1 mark

OR**Loci: Fixed Point, Isosceles Triangles, Three Non-Collinear Points** 5 marks**(a)** The locus of a point P equidistant from a fixed point O is a **circle** with centre O and radius equal to the fixed distance. 1 mark

Sketch: a circle centred at O with any radius r , where every point on the circumference is at distance r from O.

**(b)** $\triangle XAB$ and $\triangle YAB$ are isosceles with $XA = XB$ and $YA = YB$ (equal arms). 2 marks

Since $XA = XB$, X is equidistant from A and B $\Rightarrow$ X lies on the perpendicular bisector of AB.

Since $YA = YB$, Y is also equidistant from A and B $\Rightarrow$ Y also lies on the perpendicular bisector of AB.

Therefore X, Y, and their line XY all lie on the perpendicular bisector of AB, meaning XY bisects AB at right angles (perpendicularly). $\checkmark$

(c) Draw perpendicular bisectors of AB and BC. They intersect at a point O where $OA = OB = OC$. The locus is the single point O, called the **circumcentre**. 2 marks A circle drawn with O as centre and OA as radius passes through all three points and is called the **circumcircle**.